VIRTUAL PRIVATE CLOUD (VPC)

VPC (Virtual Private Cloud)

1. virtual private cloud is a virtual network that closely resembles a traditional networking that you operate in your own datacentre, with the benefits of using the scalable infrastructure of aws.
2. It is a cloud computing service provided by Amazon Web Services (AWS) that allows users to create a virtual private network (VPN) in the cloud.
3. It allows you to create a logically isolated network inside AWS where you can launch and manage your resources securely.
4. In VPC, we can define our own topology like subnets, route tables, internet gateway, Nat gateway, Security Groups and Network ACLs
5. VPC is a region-specific service so by default You can create up to 5 VPCs per region while subnet is Availability Zone (AZ) specific & we create up to 200 subnets per VPC.
6. When you create a vpc a DHCP, Network ACL, security group & route table will be automatically created.
7. Overall, a vpc is a virtual private cloud or the virtual private network that provides the security and the flexibility to host the aws resources.
8. if you need different CIDR size, create a new vpc.
9. For VPC peering connection, both VPCs must use different (non-overlapping) CIDR ranges; otherwise, peering will not work.

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Note:

DHCP

It is a dynamic host configuration protocol used to assign IP addresses to the devices.

CIDR

1. It is a classless inter domain routing used to divide the IP addresses into the flexible network sizes
2. It also used to save the IP addresses and makes routing on the internet more efficient

What is the difference between the CIDR and DHCP

CIDR allocate the IP address while the DHCP assign IP addresses to the devices.

TYPES OF VPC

1. Default VPC
2. Custom VPC

Default VPC

1. it is created default in each region when an aws account is created.
2. Has default CIDR, security group, NACL and route table settings.
3. Has an internet getaway default.

Note:

1. Default vpc range --------> 172.31.0.0/16
2. Default subnet range ---> 1. 172.31.0.0/20
   2. 172.31.16.0/20
   3. 172.31.32.0/20

Custom VPC

1. it is a vpc on aws account owner creates.
2. aws user creating the custom vpc can decide the CIDR.
3. has its own security group, network acl and route tables.
4. does not have a internet getaway by default, one needs to be created if needed.

Steps To Create a Vpc

1. create a vpc
2. create a subnets
3. create a internet gateway
4. create a route table

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Note:

1. We don't create subnets bigger than VPC, we create smaller networks inside the VPC using higher CIDR numbers.
   1. /16 = bigger network -----> 65536 hosts
   2. /20 = smaller network -----> 4096
2. As CIDR number increases, network size decreases and hosts decrease.
3. Simple to understand
   CIDR ↑ → Hosts ↓
   CIDR ↓ → Hosts ↑

4. Why do we create smaller networks (subnets) within a VPC?
    1. **To organize and manage resources easily**
       Subnets help divide a large VPC into smaller, manageable sections.
    2. **For better security**
       We can place resources in different subnets (public/private) and control access using route tables and security groups.
    3. **To isolate resources**
       For example, keep databases in private subnets and web servers in public subnets.
    4. **For high availability**
       Subnets can be created in different Availability Zones to improve fault tolerance.

We create smaller networks (subnets) inside a VPC to organize, secure, and manage resources more efficiently.

What is Subnet

1. Subnet is a range of an IP address
2. It's a network within a network

What is Subnetting

Dividing the larger network into the smaller one

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Types of Subnets

1. public subnet
2. private subnet

Public subnet

1. In public subnet the internet getway should be attach. They have known the route of internet getway.
2. if a subnet traffic is routed to an internet getway, the subnet is known as public subnet.
3. the instance in public subnet can send outbound traffic directly to the internet.
4. if you want your instance on public subnet to communicate with the internet over IPv4. It must have a public IPv4 address on elastic IP address.

Private subnet

1. In private subnet the internet getway should not be attached.

2. If a subnet does not have a route to internet getway the subnet known as private subnet.
3. the instance in a private subnet cannot send outbound traffic directly to the internet.
4. There are basically 5 IP addresses that reserve by the aws

    1) 10.0.0.0 ----------> network address
    2) 10.0.0.1 ----------> vpc route
    3) 10.0.0.2 ----------> DNS server
    4) 10.0.0.3 ----------> future use
    5) 10.0.0.255 -------> broadcast address

Note:

1. aws does not support broadcast in a vpc but reserve these 5 IP addresses.
2. As well as we can get only the IP address range from /16 to /28

INTERNET GATEWAY

1. it is kind of router.
2. it also a virtual router that connects VPC to the internet.
3. if you create a new VPC then you must attach the internet getway in order to access internet.
4. ensure that u must add the internet gateway in the route table so that these subnets can also route to the internet.
5. it performs NAT between your private and public IPv4 address.
    a. Converts private IP → public IP for outgoing traffic
    b. Converts public IP → private IP for incoming responses
6. it supports both IPv4 and IPv6.
7. Default route example for Internet access:
    a. 0.0.0.0/0 → Internet Gateway (for IPv4)
    b. ::/0 → Internet Gateway (for IPv6)

ROUTE TABLE

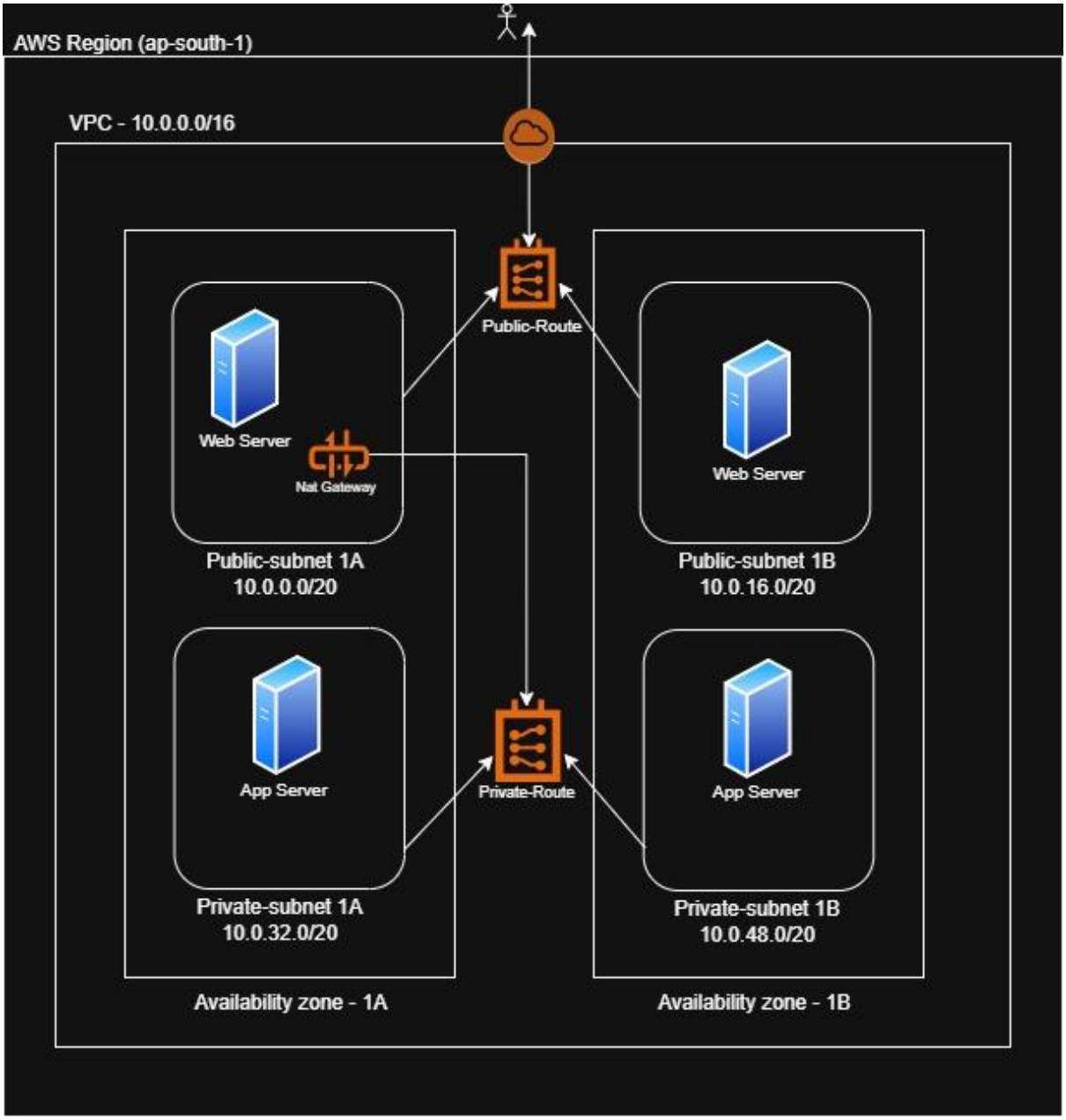
1. it is a central routing function.
2. it connects the different availability zone together and connects VPC to the internet getway.
3. you can have upto 200 routes table per vpc.
4. you can have upto 50 routes entries per route table.
5. each subnet must be associated with only one route table at any given time.

6. if you do not specify a subnet to route table association, the subnet will be associates with the default VPC route table
7. you can also edit the main route table if you need, but you cannot delete main route table.
8. however, you can make a custom route table manually become the main route table then you can delete the former main, as it is no longer a main route table.
9. you can associate multiple subnets with the same route table.

CREATE A FIRST VPC

1. **Create A Vpc**
    1. name --------> myvpc
    2. cidr range --> (10.0.0.0/16)
2. **Create A Subnet**
    a) name ----------> public subnet
    b) vpc ----------> myvpc
    c) subnet range ---> 10.0.0.0/24
3. **Create A Internet Gateway & Attach To A Vpc**
4. **Create A Route-Table**
    a) Public-route-table
    1. associates public subnets
    2. attach Internet gateway (0.0.0.0/0)
5. **Create A Instances**
    1. name ----------> public-server
    2. ami ----------→ ubuntu
    3. key pair ---------> ubuntu.pem
    4. instance type --> t3.micro
    5. vpc ----------> myvpc
    6. subnet ----------> public subnet
    7. create a instance
6. **Connect The Instance**
    1. connect the public server
    2. ping 8.8.8.8

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AWS Region (ap-south-1)
VPC - 10.0.0.0/16
Public-Route
Web Server
Nat Gateway
Web Server
Public-subnet 1A
10.0.0.0/20
Public-subnet 1B
10.0.16.0/20
App Server
Private-Route
App Server
Private-subnet 1A
10.0.32.0/20
Private-subnet 1B
10.0.48.0/20
Availability zone - 1A
Availability zone - 1B